

Imperatae rhizoma-Hedyotis diffusa Willd. herbal pair alleviates nephrotic syndrome by integrating anti-inflammatory and hypolipidaemic effects

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ABSTRACT

Background: Nephrotic syndrome (NS) is a common nephropathy with a complex and diverse aetiology. Both *Imperatae rhizoma* and *Hedyotis diffusa* Willd. are herbs that are widely used as medicine and functional food. In traditional Chinese medicine theory, they are used as an herbal pair (HP) to treat inflammation-related diseases in the clinic, especially disorders of the kidney.

Purpose: This study aimed to investigate the anti-inflammatory and hypolipidaemic effects of HP in an NS rat model and provide scientific data for its clinical application.

Methods: An NS model was established by two-dose injection of Sprague-Dawley rats with adriamycin. Seven groups, including the sham, model, HP treatment (0.25, 0.5 and 1.0 g/kg/d), prednisone (positive control, 5 mg/kg/d), and atorvastatin (positive control, 4 mg/kg/d) groups, were tested. The biochemical indexes of renal function and inflammatory cytokines were determined by ELISA kits and/or qPCR assays, and the crucial protein involved in the signalling pathway were subsequently tested by qPCR and/or Western blotting. Based on specific compounds identified by LC-Q-TOF-MS, network pharmacological study was carried out.

Results: The levels of BUN, Scr, Upro, UA, Alb, TC, TG, and LDL-C were significantly elevated in model rats. HP treatment for four weeks improved the renal function and the dyslipidaemia by decreasing the levels of all parameters, except BUN and Scr. HP treatment (0.5 and 1.0 g/kg/d) upregulated the expression of PPARγ, CYP7b1, and LDLR in the liver, while it down-regulated PCSK9, showing a regulatory effect on lipid metabolism disorder. The levels of TNF-α and IL-1β in the plasma and the mRNA expression of *TNF-α*, *IL-1β*, *MCP-1*, and *TGF-β1* in the kidney were decreased in HP groups, revealing its anti-inflammatory effect in NS rats. The HP exerted an alleviation effect on the inflammatory response through the NF-κB pathway by inhibiting the mRNA and protein expression of p50 and p65. There were 34 compounds identified or tentatively characterized in HP. In the network pharmacological study, PPARG(PPARγ), PCSK9, RELA(p65), and NF-κB1(p50) were the top 20 targets for HP, supporting the animal experimental results.

Abbreviations: Alb, albumin; BUN, urea nitrogen; Cyp7a1, cholesterol 7 alpha hydroxylase; Cyp7b1, oxysterol 7α-hydroxylase; Cyp8b1, cholesterol 12α-hydroxylase; Cyp27a1, sterol 27-hydroxylase; ERK, extracellular signal-regulated kinase; GO, Gene Ontology; H&E staining, hematoxylin and eosin; HDL, high-density lipoprotein; HP, *Imperatae rhizoma* and *Hedyotis diffusa* Willd. herbal pair; JNK, c-Jun N-terminal kinase; KEGG, Kyoto Encyclopedia of Genes and Genomes; MCP-1, monocyte chemoattractant protein-1; LDLR, Low density lipoprotein receptor; LDH, lactate dehydrogenase; IL-1β, interleukin-1β; IL-1R, interleukin-1 receptor; LC-Q-TOF-MS, liquid chromatography coupled to quadrupole time-of-flight mass spectrometry; NF-κB, nuclear factor-kappa B; NS, nephrotic syndrome; ORO staining, oil red O staining; PCSK9, proprotein convertase subtilisin/kexin type 9; PPARα, peroxisome proliferator-activated receptor α; PPARγ, peroxisome proliferator-activated receptor-γ; PVDF, poly-vinylidene fluoride; qPCR, quantitative real time PCR; SDS-PAGE, sodium dodecyl sulfate-polyacrylamide gel electrophoresis; SREBP2, sterol regulatory element binding protein 2; TC, total cholesterol; TG, triglyceride; TGF-β1, transforming growth factor-β1; TNF-α, tumor necrosis factor-α; TNFR, tumor necrosis factor receptor; UA, uric acid; Upro, urinary protein.

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Conclusion: : HP exhibited protective effects on NS rats. These effects might be closely related to the inhibition of NF- κ B and PCSK9-LDLR and activation of the PPAR γ -CYP7B1 signalling pathways.

Introduction

Nephrotic syndrome (NS) is a highly prevalent kidney disease characterized by hyperalbuminuria (≥ 3.5 g/24 h), oedema, hyperlipidaemia, and hypoproteinaemia in the clinic (Macé and Chugh, 2014). Although most patients respond to the initial therapy with glucocorticoids by entering clinical remission, approximately 10–20% of children and 50% of adults either present with or subsequently progress to steroid-resistant NS (SRNS), and approximately 50% of steroid-sensitive patients will suffer frequently relapsing NS (FRNS) or steroid-dependent NS (SDNS) during the course of this disease (Fu et al., 2017; McKinney et al., 2001). Failure to achieve complete health will greatly increase the risk of various complications, such as atherosclerosis and thromboembolism, which are closely associated with the magnitude of long-term hyperlipidaemia (Agrawal et al., 2018).

Both *Imperatae rhizoma* and *Hedyotis diffusa* Willd. are prescribed in the Guidelines for the Treatment of Chronic Kidney Disease recommended by the China Association of Chinese Medicine (China Association of Chinese Medicine, 2011, 2019). *I. rhizoma* and *H. diffusa* herbal pair (HP) was a clinical prescription for the treatment of nephropathy based on the traditional Chinese medicine theory that *I. rhizoma* has the function of clearing heat, cooling blood, and inducing diuresis for removing oedema, while *H. diffusa* can clear heat-toxicity; thus, they could well supplement and help each other (Liu and Li 2010; Wang and Zhang, 2017), and this HP has been used for more than thirty years (Wang, 1987). Meanwhile, these two herbs are essential components in Shenyang Kangfu Pian, which was recorded in the Chinese Pharmacopoeia (2020 edition) for the treatment of renal disease (Chinese Pharmacopoeia Commission, 2020). *I. rhizoma* (family Gramineae) is commonly used in traditional Chinese medicine and was originally described in Shennong's Classic of Material Medical. It contains triterpenoids, saccharides, organic acids, and flavonoids, which perform positive pharmacological roles in antioxidant, anti-complementary, and immunomodulatory activities (Fu et al., 2010; Lu and Huang, 1996; Zhou et al., 2013). Modern pharmacological studies have proven that saccharides are the main active component of *I. rhizoma* with hypoglycaemic action and regulation of lipid metabolism effects (Cui et al. 2012). *H. diffusa* (Family Rubiaceae) is a well-known ethnic medicine with anti-inflammatory, antitumor, and antifibrosis activities, as well as immunomodulatory and neuroprotective effects (Chen et al., 2016). We have performed studies on *H. diffusa*, including chemical analysis, pharmacological actions, and network pharmacology (He et al, 2018a, 2018b; Lu et al., 2019; Ye et al., 2015), and demonstrated that *H. diffusa* has a remarkable anti-inflammatory effect in the treatment of kidney-related diseases. Inflammation and lipid metabolism disorders are two principal aspects of NS. Hence, we propose a hypothesis that, supplementing an anti-inflammatory effect of *H. diffusa* with a regulating lipid metabolism effect of *I. rhizoma*, HP could exert a good pharmacological effect on NS through these two aspects.

Before this study, the biological effects of *H. diffusa*, *I. rhizoma*, and HP on NS rats were preliminarily tested with the method described in this manuscript. The results, presented in Table S1, showed that HP showed a more comprehensive and desirable pharmacological effect than the two individual herbs. Then, this study aims to investigate the pharmacological effects of HP on NS, focusing particularly on revealing the mechanisms of its anti-inflammatory and hypolipidaemic activities, to provide data for its clinical application.

Materials and methods

Chemicals and reagents

The herbs *I. rhizoma* and *H. diffusa* were purchased from Anhui Fuzhong Pharmaceutical Co. (Bozhou, China). Asperulosidic acid, asperuloside, desacetyl asperulosidic acid, scandoside methyl ester, and *E*-6-*O*-*p*-coumaroyl scandoside methyl ester (purity $\geq 98\%$, all) were purchased from Shanghai Yuanye Biotechnology Co., Ltd. (Shanghai, China). Chlorogenic acid (purity $\geq 98\%$) was purchased from the National Institute for the Control of Pharmaceutical and Biological Products (Beijing, China). Prednisone and atorvastatin (purity $\geq 98\%$, both) were purchased from Dalian Meilunbio® Co., Ltd (Dalian, China). Adriamycin was purchased from Anhui Kuer Biological Engineering Co. Ltd (Hefei, China). The kits for analysis of blood urea nitrogen (BUN), uric acid (UA), albumin (Alb), urinary protein (Upro), lactate dehydrogenase (LDH), total cholesterol (TC) and triglyceride (TG) were purchased from Nanjing Jiancheng Bioengineering Institute (Nanjing, China). The ELISA kits for IL-1 β and TNF- α were purchased from Tsingmu Biotechnology (Shenzhen) Co., Ltd. (Shenzhen, China). HPLC grade acetonitrile was purchased from Fisher Scientific (Fair Lawn, NJ, USA). Antibodies of p50, p65, PCSK9, LDLR and GAPDH were purchased from Beijing Bioss Biotechnology Co., Ltd (Beijing, China). Highly pure water was produced by Milli-Q Academic (Bedford, MA, USA). Formic acid (HPLC-grade) was purchased from Sigma-Aldrich (St Louis, MO, USA).

HP aqueous solution preparation

HPs were prepared using the water decocting method according to Shenyang Kangfu Pian and further optimized. The dried mixture of *I. rhizoma* and *H. diffusa* (crude drug ratio 1:1) was soaked in 10 times distilled water for 30 min and decocted for 1 h. After filtration through gauze, the residue was decocted with 10 times distilled water for 1 h and then with 5 times distilled water for 0.5 h. Three filtrates were combined and evaporated to a concentration of 0.2 g crude drugs/mL under reduced pressure at 60 °C. The obtained solution was stored at –20 °C for subsequent experiments.

Animal experiments

Seven-week-old male Sprague-Dawley (SD) rats (200–220 g body weight) were obtained from the Laboratory Animal Center of Southern Medical University (Guangzhou, China; Certification: SCXK-Yue-2016-0041). All experiments were performed according to the Regulations on the Administration of Experimental Animals of Southern Medical University, Laboratory Animal Guideline for Ethical Review of Animal Welfare (GB/T 35892-2018) and Laboratory Animal Requirements of Environment and Housing Facilities (GB 14925-2010) and were approved by the Animal Ethics Committee of School of Southern Medical University on 20/12/2018. All animals were housed in an air-conditioned room maintained at 23 $\pm$ 2 °C, 40 $\pm$ 5% relative humidity and an alternating 12 h dark/light cycle. Animals had food and water *ad libitum* throughout the experiment.

Six sham and 36 model rats were used in the experiment. The model was established by caudal vein injection of adriamycin once a week for two weeks. The dose was 4 mg/kg for the first week and 2 mg/kg for the second week. Then, they were randomly divided into 7 groups (each group $n = 6$): sham group, NS group, HP treatment groups (0.25, 0.5 and 1.0 g/kg/d), prednisone group (positive control, 5 mg/kg/d) and

atorvastatin group (positive control, 4 mg/kg/d). The drug was given by oral administration once a day for four consecutive weeks. The sham group was given the same amount of normal saline throughout the experiment. The weight of each group was recorded in the 2nd, 4th and 6th weeks. Urine was collected on the 41st day in a metabolic cage for a period of 24 h to detect the content of Upro according to the manufacturer's instructions. Serum was collected on the 42nd day from the orbital vein of rats for the determination of BUN, Scr, UA, Alb, TC, TG, LDL-C, TNF- α and IL-1 β as per the manufacturers' instructions for commercial ELISA kits. Finally, all rats were anaesthetized with 10% chloral hydrate intraperitoneally at a dose of 0.3 ml/100 g and sacrificed by cervical dislocation (Lu et al., 2019). The liver and kidney tissues were weighed. The right kidneys of all groups were cut into two parts. One part was fixed in 4% formalin for haematoxylin and eosin (H&E) staining and Oil red O (ORO) staining according to our previous method (Lu et al., 2019; Zhang et al., 2021), and the other was stored in liquid nitrogen for the detection of RNA or protein.

RNA reverse transcription and quantitative real-time PCR (qPCR)

Total RNA was extracted from the liver/renal tissue using an RNAprep Pure Tissue Kit according to the manufacturer's protocol (TIANGEN BIOTECH (Beijing) Co., LTD, China). The reverse transcription system (cDNA Synthesis) contained 500 ng total RNA, 2 μ l 5 $\times$ PrimeScript RT Master Mix (TaKaRa, PrimeScriptTM RT Master Mix, Guangzhou, China) and 10 μ l RNase-free ddH₂O. The mixed sample was then placed into an ETC811 PCR amplifier (Beijing Dongsheng Innovation Biotechnology Co., Ltd, China) for reverse transcription with the following reaction conditions: incubation at 37 °C for 15 min and incubation at 85 °C for 5 s. The real-time PCR system contained 10 μ l 2 $\times$ PCR Master Mix, 0.4 μ l PCR Forward Primer (10 μ M), 0.4 μ l PCR Reverse Primer (10 μ M), 7.2 μ l RNase Free H₂O, and 2.0 μ l cDNA. It was carried out on a Roche LightCycler® 480 real-time PCR instrument (Roche, Basel, Switzerland) under the following reaction conditions: 95.0 °C for 2 min, followed by 40 cycles of 95.0 °C for 15 s and 60.0 °C for 1 min. The primer sequences of the genes are shown in Table 1. The threshold cycle (C_t) was recorded by LightCycler 480 Software (release 1.5.0), and the relative fold change was calculated using the $\Delta\Delta C_t$ method ($2^{-\Delta\Delta C_t}$).

Western blot

The proposed protein extraction of liver/kidney tissues was performed according to our previous reports (Lu et al., 2019). The protein concentration was measured by a BCA protein assay kit (Beyotime Biotechnology, Shanghai, China) following the manufacturer's instructions. After protein denaturation, a certain amount of protein and molecular weight marker were loaded into the wells of 10% sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) for 30 min, and then the gel was run for 75 min at 120 V. Subsequently, the

protein was transferred to a polyvinylidene fluoride (PVDF) membrane and run for 90 min at 100 V. After blocking for 2 h at 4 °C, the membrane was incubated overnight with primary antibodies against p50 (1:1000), p65 (1:1000), PSCK9 (1:1000), LDLR (1:1000), GAPDH (1:10000) and β -actin (1:10000) at 4 °C and then washed three times. The membrane was incubated with the secondary antibody (1:10,000) at room temperature for 2 h and washed three times for 5 min each. Finally, the blots were scanned using a Fluorchem R multifunctional imaging analysis system (ProteinSimple, USA). Semiquantitative analysis was performed using ImageJ software (<https://ij.imjoy.io/>) by measuring the intensity in three experiments.

Chemical profile of HP aqueous solution

The chemical profile of HP was analyzed by an LC-Q-TOF-MS system consisting of an Agilent 1290 LC system and an Agilent 6540 Q-TOF-MS system (Agilent Technology, Santa Clara, CA, USA). Experiments were carried out on an Agilent ZORBAX Eclipse XDB-C18 (3.0 $\times$ 150 mm i.d., 3.5 Micron) for chromatographic separation. The mobile phase was composed of acetonitrile (A) and 0.1% formic acid (B) at a flow rate of 0.5 ml/min. The following gradient elution procedure was started with 5% B for 5 min and changed linearly to 60% B from 5 min to 30 min with the column temperature at 25 °C. Both positive and negative ion modes were used in MS detection, with the following parameters: Vcap 3500 V, nozzle voltage 1000 V, fragmentor 175 V, skimmer 1 65 V, gas temperature 250 °C, gas flow 14 L/min, nebulizer 35 psig, and sheath gas temperature 350 °C. The *m/z* scan range was 100–1000. The structures of these molecules were obtained from the NCBI PubChem database (<http://www.ncbi.nlm.nih.gov/pccompound>), ChemSpider (<http://www.chemspider.com/>) and ZINC (<http://zinc.docking.org>) and saved in mol2 format for further analysis.

In addition, the contents of deacetylcholine acid and chlorogenic acid were quantified for HP quality control on an Agilent 1290 LC system with a C18 column (3.0 mm $\times$ 100 mm i.d, 2.7 Micron) at room temperature. The mobile phase was a binary eluent of 0.1% formic acid solution (v/v) (A) and acetonitrile (B) with the following gradient conditions: 0–5 min, 2% B, 5–40 min, 2–10% B. The injection volume was 10 μ l, and the flow rate was 0.3 ml/min. The UV detection wavelength was 235 nm.

Network pharmacological process

The candidate disease targets for NS were screened through network databases, including DisGeNET (<https://www.disgenet.org/>), OMIM (<https://omim.org/>), DiGSeE (<http://210.107.182.61/geneSearch/>), and MalaCards (<https://www.malacards.org/>), by inputting the keyword “Nephrotic Syndrome”. Their corresponding X-ray crystallographic structures were directly downloaded from the RCSB PDB database (<http://www1.rcsb.org/>) and subsequently propagated by Sybyl 7.3 software. Molecular docking for chemical compounds and NS-

Table 1
Primer sequences for quantitative real-time PCR amplification.

Name	Primer-Forward	Primer-Reverse	Ref.
<i>Cyp7a1</i>	ACGCACCTCGCTATTCTCTG	GGCAGGTCATTCAAGTTGCAC	(Zhu et al., 2013)
<i>Cyp7b1</i>	TGACAGGATGAGAGAGAGAGA	TAGAGCCACCAATCCACACA	(Zhu et al., 2013)
<i>Cyp8b1</i>	GTACACATGGACCCGACATC	GGGTGCCATCAGGGTTGAG	(Zhu et al., 2013)
<i>Cyp27a1</i>	ACCTTTCTGAGCTGATCTTGGCT	GCATGTGGGCAAGTCCTTGTCT	(Zhu et al., 2013)
<i>HMG-CoA</i>	GGTGGTGGGACCAACCTTCT	CACGCCCTTGAACACCTA	(Wa et al., 2019)
<i>LDLR</i>	TAGGGTTTCTGCTTTCACCA	GCCACCACATTCTTCAGGTT	(Ding et al., 2018)
<i>IL-1β</i>	CACCTCTCAAGCAGAGCACAG	GGGTTCATGGTGAAGTCAAC	(Wei et al., 2019)
<i>MCP-1</i>	GCCTGCTGTTTACAGTTGC	CAGGTGAGTGGGGCGTTA	(Wei et al., 2019)
<i>p50</i>	TTCAACATGGCAGACGACGA	AGGTATGGGCGATCTGTTGAC	(Lu et al., 2019)
<i>p65</i>	TGCAGGCTCCTGTGCGAGTG	TCCGGTGGCGATCGTCTGTGT	(Lu et al., 2019)
<i>PCSK9</i>	TGGCTGCATGACATTGCTTCTC	GCACTGGAGAACCACACAGG	(Sujatys-Szulc et al., 2016)
<i>SREBP2</i>	GATGAGCTGACTCTCGGGACATC	GTGGGTAGGAGAGACTTTGACCT	(Han et al., 2015)
<i>TNF-α</i>	CCAGGTTCTCTTCAAGGGACAA	CTCCTGGTATGAAATGGCAATC	(Wei et al., 2019)

related disease targets was performed by the Docking Suite program, which is consistent with what we reported previously (Dong et al., 2020; Gong et al., 2021). A docking score ≥ 6.0 was taken as the screening condition for active compounds. Subsequently, the compound-target network was obtained by importing active compounds and related targets into Cytoscape 3.7.1 software (<https://cytoscape.org/>). For the illustration of the interactions between proteins, the potential targets were uploaded to the online software String11.0 (<https://string-db.org/>), and the interaction value with a score of > 0.7 was regarded as high confidence data. The obtained protein interaction data were imported into Cytoscape 3.7.1 software to construct a protein-protein interaction (PPI) network, and the bar chart of high-salience targets in the PPI network was drawn in Excel. To clarify the role of the targets in gene function and signalling pathways, the David V 6.8 database (<https://david.ncifcrf.gov/>) was used to carry out Gene Ontology (GO) function enrichment and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analyses. The obtained GO function enrichment bubble diagram was drawn using the YHBIO mapping tool (<http://www.ehbio.com/ImageGP/>), and a target-pathway network was constructed using Cytoscape 3.7.1 software.

Statistical analysis

All data were analyzed using GraphPad Prism (GraphPad Software, San Diego, CA) and are shown as the mean $\pm$ standard deviation (SD). SPSS 20.0 statistical software (SPSS Inc. Chicago, USA) was employed to test for significant differences between groups ($p < 0.05$, $p < 0.01$, $p < 0.001$) by one-way ANOVA.

Results

Effects of HP on the body weight of adriamycin-induced NS rats

Body weight was measured every two weeks throughout the experimental period. After adriamycin injection, diet and water intake in the NS group were obviously reduced, which was fully reflected in the weight. As shown in Table 2, the body weight of rats treated with adriamycin significantly decreased in comparison with that of the sham group in the 2nd week. In comparison with the NS group, treatment with HP, prednisone and atorvastatin prevented the loss of body weight in adriamycin-induced NS rats at the 4th and 6th weeks.

Effects of HP on the biochemical parameters of adriamycin-induced NS rats

The characteristics of NS in rats include oedema, increased proteinuria and hyperlipidaemia, and decreased Alb. The kidney index was calculated as the ratio of the weight of the right kidney to the body weight on the 6th week. In comparison with the sham group, the right renal index of the NS model group increased by 1.71 times (Fig. 1A). After HP administration for 4 weeks, the kidney indexes were reduced

by 3%, 12%, and 18% in the three different dose groups (0.25, 0.5, and 1.0 g/kg), respectively. For the positive control drugs, the renal indexes were reduced by 12% and 17% in the prednisone and atorvastatin groups, respectively. It is worth noting that degree of decrease in the renal index was similar between the high HP dose group and atorvastatin group. The levels of Upro and UA were significantly increased in NS model rats, but no significant differences are found in the levels of BUN and Scr (Fig. S1). After 4 weeks of treatment, the low dose (0.25 g/kg) and medium dose (0.5 g/kg) of HP showed a faint relieving effect on the level of Upro, while it was reduced by 25% ($p < 0.05$) in the high dose (1.0 g/kg) group (Fig. 1B). The levels of UA in the three HP groups (0.25, 0.5, and 1.0 g/kg) were all notably decreased by 31%, 37%, and 54% ($p < 0.05$), respectively (Fig. 1C). The Alb level of NS rats was significantly decreased, but HP treatment (0.25, 0.5, and 1.0 g/kg) increased Alb levels in a dose-dependent manner (Fig. 1D). Compared with the NS model group, TC, TG and LDL-C were all reduced by 30% - 41%, 34% - 39%, and 53% - 63% after HP oral administration at three doses of 0.25, 0.5, and 1.0 g/kg, respectively, while atorvastatin decreased the TG and LDL-C levels (Fig. 1E-G). Meanwhile, the levels of TC and TG in renal tissues of rats in the middle and high HP groups (0.5 and 1.0 g/kg) were also significantly decreased (Fig. 1H-I). It is worth noting that the TG level was increased in the prednisone group, which might be related to the disorder of glucose and lipid metabolism induced by long-term use of glucocorticoids (Fardet and Fève, 2014). These results indicated that HP has a remarkable regulatory effect on hyperlipidaemia in adriamycin-induced NS rats.

Effects of HP on inflammatory factors in adriamycin-induced NS rats

Anti-inflammation is an important treatment of NS. As shown in Fig. 2(A, B), the levels of the inflammatory cytokines TNF- α and IL-1 β in NS rats were 5.46 times and 7.25 times higher than those in control rats, respectively. The contents of TNF- α and IL-1 β in the HP groups (0.25, 0.5 and 1.0 g/ml) were decreased in a dose-dependent manner. Remarkably, the anti-inflammatory effect of the high-dose group (1.0 g/ml) was slightly lower than that of the prednisone group, and the anti-inflammatory effect of atorvastatin was similar to that of the medium-dose group. The mRNA expression of TNF- α and IL-1 β in the kidney showed the same trend as that in the serum. Moreover, the mRNA expression of the other two inflammatory factors, MCP-1 and TGF- β 1, was also measured (Fig. 2C). Compared with those in the sham group, the mRNA levels of MCP-1 and TGF- β 1 in the kidneys of the NS group were increased by 10.0- and 2.56-fold, respectively, and they were significantly downregulated in the HP groups at doses of 0.5 and 1.0 g/kg. In addition, the mRNA levels of IL-1 β , TNF- α , MCP-1 and TGF- β 1 in the HP group (1.0 g/kg) were not significantly different from those in the prednisone group.

Effect of HP on pathological changes of kidney

To verify the effect of HP, pathological changes in the kidney were

Table 2
The body weight of every group.

Groups	Pre-modeling	The 2 nd week	The 4 th week	The 6 th week
sham	237.83 $\pm$ 13.73	269.83 $\pm$ 10.09	346.1 $\pm$ 14.39	362.1 $\pm$ 9.48
NS model	243.99 $\pm$ 8.75	218.50 $\pm$ 10.70###	200.44 $\pm$ 9.79###	195.67 $\pm$ 10.17###
HP 0.25 g/kg	234.67 $\pm$ 13.54	222.00 $\pm$ 12.82###	258.17 $\pm$ 14.40***	258.37 $\pm$ 12.99***
HP 0.5 g/kg	237.17 $\pm$ 10.02	224.50 $\pm$ 15.76###	281.76 $\pm$ 12.74***	294.45 $\pm$ 15.76***
HP 1.0 g/kg	235.17 $\pm$ 9.05	218.01 $\pm$ 4.63###	289.64 $\pm$ 12.24***	292.68 $\pm$ 11.07***
Prednisone (5 mg/kg)	243.00 $\pm$ 7.30	229.33 $\pm$ 3.54###	255.49 $\pm$ 6.01***	241.17 $\pm$ 13.02***
Atorvastatin (4 mg/kg)	238.83 $\pm$ 7.82	230.00 $\pm$ 13.05###	250.25 $\pm$ 11.15***	272.95 $\pm$ 9.70***

Data are presented as mean $\pm$ SD (n = 6). Compared with sham group, # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$; Compared with NS group, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

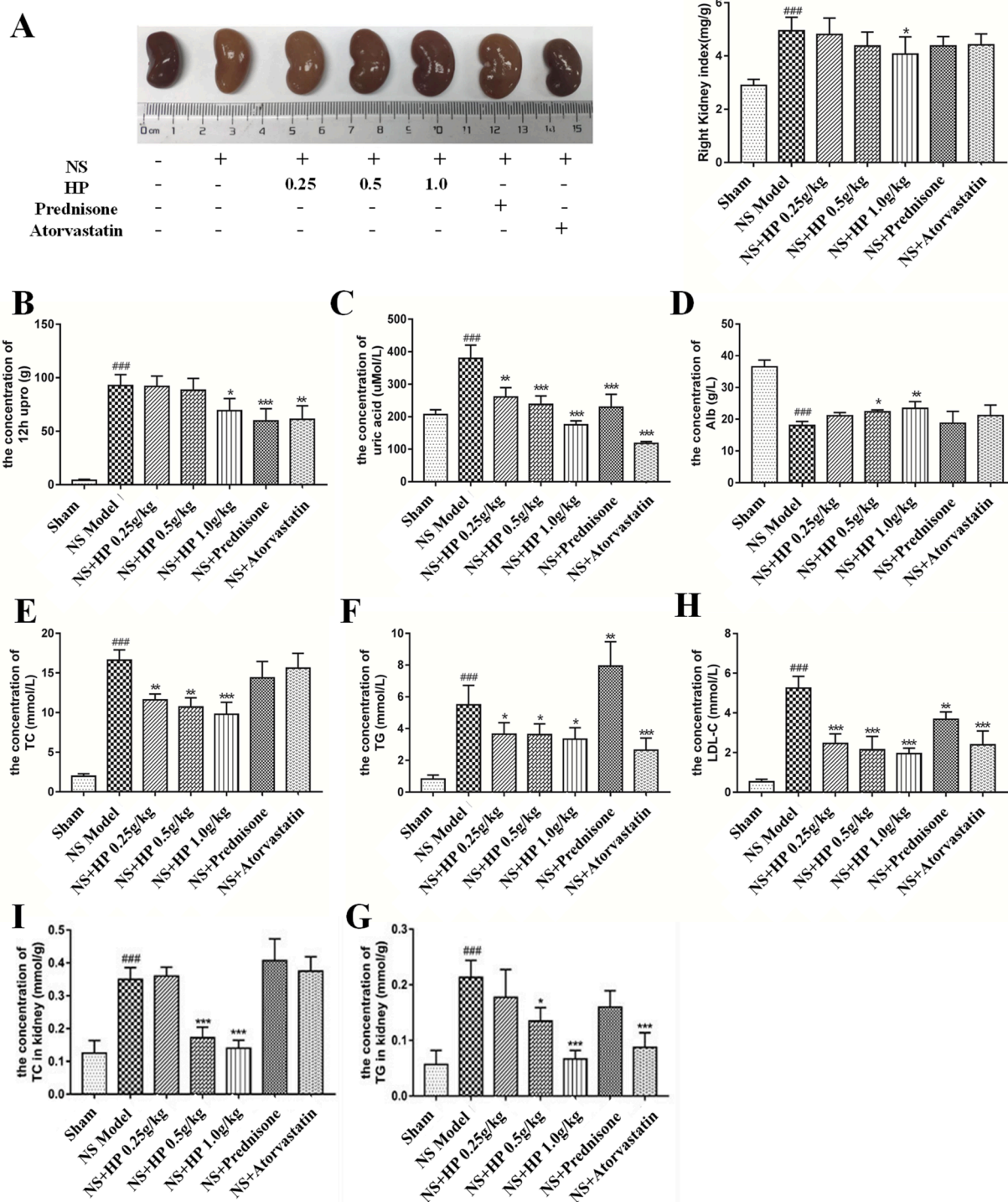


Fig. 1. Effect of HP on the biochemical parameters in NS rats. The kidney index was calculated as the ratio of the weight of the right kidney to the body weight on the 6th week after adriamycin injection (A). UA (B) and Upro (C) content at 24 h were measured on the 4th week after treatment, respectively. Alb (D), TC (E), TG (F) and LDL-C (G) in serum were determined on the 4th week after treatment, as well as TC (H), TG (I) in kidney tissue. Data are presented as mean $\pm$ SD ($n = 6$). Compared with sham group, # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$; Compared with NS group, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

observed by H&E and ORO staining. After adriamycin injection for 6 weeks, renal oedema was obviously observed, and the colour became lighter in comparison with the sham group. As shown in the H&E staining of renal sections (Fig. 3A), a thickened glomerular basement membrane, glomerular sclerotic lesions, focal interstitial haemorrhaging, and vacuoles in some renal tubule epithelial cells were obviously observed in the NS group. However, this pathological damage was alleviated by HP treatment for 4 weeks. ORO staining was performed to

assess the extent of lipid accumulation in the kidney. As shown in Fig. 3B, lipid accumulation was obviously observed in the NS group compared with the sham group. The kidneys of rats in the HP, prednisone and atorvastatin groups were lighter red, and lipid accumulation was improved.

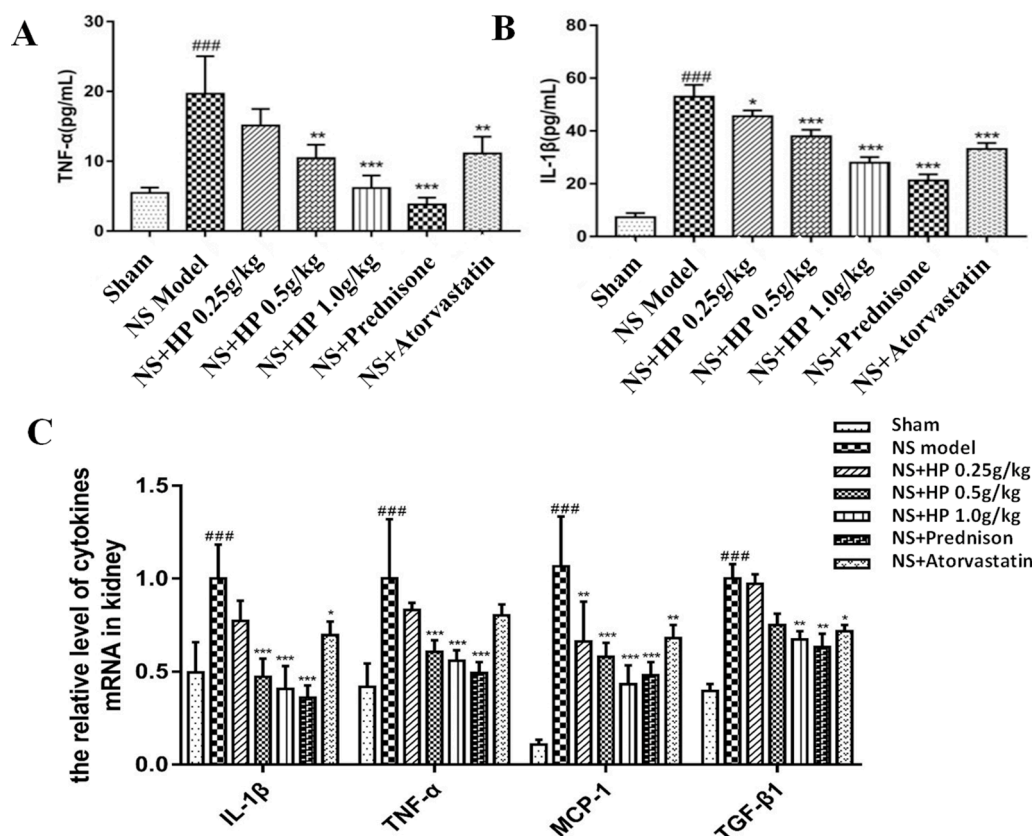


Fig. 2. Effect of HP on inflammatory cytokines in NS rats. The contents of TNF- α (A) and IL-1 β (B) were determined after HP treated for 4 weeks in serum. The mRNA expressions of IL-1 β , TNF- α , MCP-1 and TGF- β 1 were determined in kidney tissue (C). Data are presented as mean $\pm$ SD ($n = 6$). Compared with sham group, # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$; Compared with NS group, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Effects of HP on the NF- κ B pathways in adriamycin-induced NS rats

NF- κ B plays an important role in inflammation and is involved in the synthesis of IL-1 β and TNF- α . Thus, key factors, such as I κ B- α , p50, and p65, were measured using qPCR to clarify whether HP inhibited activation of the NF- κ B signalling pathway in kidney tissue. As shown in Fig. 4, I κ B- α was significantly downregulated in NS rats but increased by 1.3- and 1.7-fold after HP treatment at 0.5 and 1.0 g/kg, respectively. The expression of p50 and p65 in the NS group was significantly upregulated ($p < 0.001$). Expression of p50 was reduced by 25%, 46% and 59% after HP treatment at 0.25, 0.5 and 1.0 g/kg, respectively, and that of p65 was decreased by 25% and 40% after HP treatment at 0.5 and 1.0 g/kg, respectively. Western blot analysis also proved that the levels of p50 and p65 in the nucleus were obviously increased with HP treatment at 1.0 g/kg. Together, the inhibition of cytokine production mentioned above indicated that HP treatment could inhibit the NF- κ B signalling pathway to alleviate the inflammatory reaction of NS rats.

Regulation effect of HP on bile acid synthesis pathway in adriamycin-induced NS rats

Bile acid synthesis disorder is one of the causes of hyperlipidaemia because bile acid biosynthesis is the main metabolic pathway of cholesterol in the body. Here, the mRNA expression levels of three enzymes (Cyp7a1, Cyp8b1 and Cyp27a1) in classical pathways and two enzymes (Cyp27a1 and Cyp7b1) in nonclassical pathways for bile acid synthesis were measured, as well as their upstream regulators PPAR α and PPAR γ in liver tissue (Fig. 5). In comparison with the sham group, the mRNA expression levels of Cyp7b1, Cyp8b1 and Cyp7a1 in NS rats were significantly decreased, while no significant change was observed

in the expression of Cyp27a1. Unfortunately, after oral HP decoction administration, the mRNA expression levels of Cyp7a1, Cyp8b1 and Cyp27a1 were not significantly different compared with those in the NS model group. However, we found that HP treatment at doses of 0.5 and 1.0 g/kg could significantly upregulate the mRNA expression of Cyp7b1 by 1.92 times and 2.06 times, respectively. In addition, the mRNA expression levels of PPAR α and PPAR γ were decreased in the NS model. HP treatment upregulated the expression of PPAR γ but not PPAR α , and the change trend was consistent with that of Cyp7b1. These results indicated that HP decoction might increase the conversion of cholesterol to bile acid through the PPAR γ -CYP7B1 pathway, thereby reducing cholesterol levels in the body.

Regulatory effect of HP on the PCSK9-LDLR pathway in adriamycin-induced NS rats

Low-density lipoprotein receptor (LDLR) is the key step of LDL clearance and can be regulated by sterol regulatory element binding protein 2 (SREBP2) and proprotein convertase subtilisin/kexin type 9 (PCSK9). As shown in Fig. 6A, PCSK9 mRNA expression in the liver of NS rats was significantly increased, compared with that of the sham group, while LDLR mRNA levels were significantly decreased. After HP (0.25, 0.5 and 1.0 g/kg) treatment, the mRNA expression of PCSK9 was decreased by 39%, 48%, and 55%, and the mRNA expression of LDLR was upregulated by 11%, 19%, and 26%, respectively. Results from Western blot test also showed the changes of PCSK9 and LDLR in the five groups, which was consistent with that of mRNA test (Fig. 6). Furthermore, the mRNA expression and protein level of SREBP2 were also tested (Fig. S2), and HP administration could not change the level of SREBP2. All the results indicated that HP has a the hypolipidaemic effect through the PCSK9-LDLR pathway rather than through SREBP2.

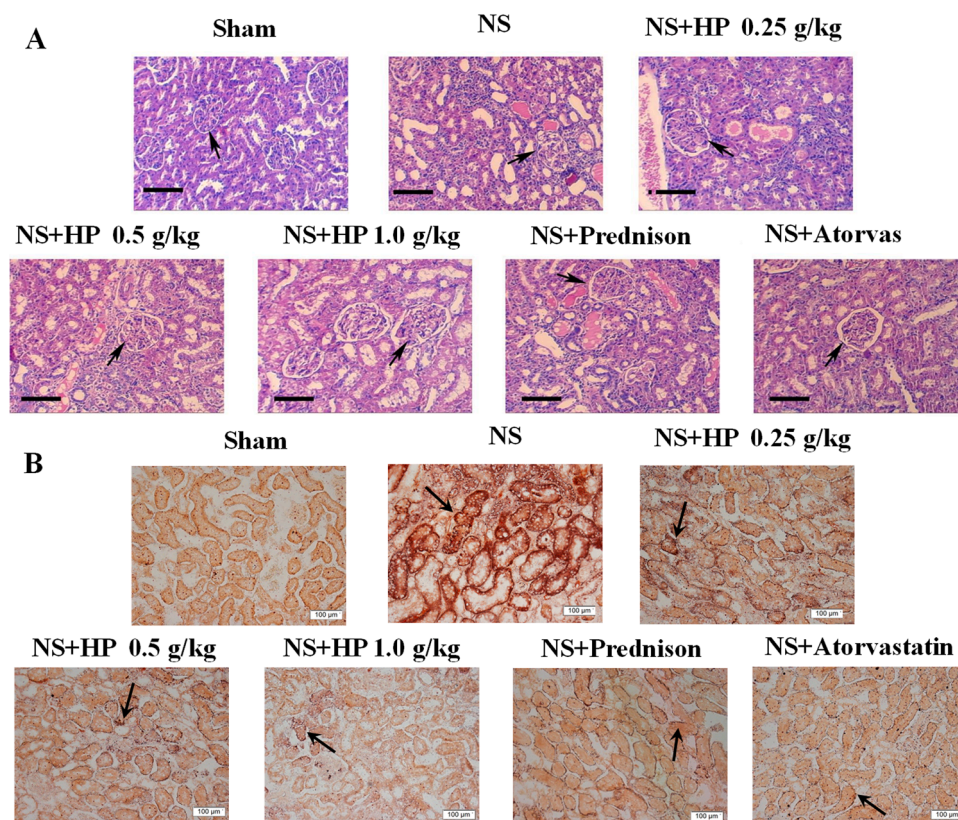


Fig. 3. Effects of HP on renal histopathological alterations in NS rats. Photomicrographs of HE stained kidney sections from each groups at the 6 th week after adriamycin injection. Scale bars: 600 μm ($\times 200$) (A); Photomicrographs of oil red stained kidney sections from the each groups at the 6 th week after Adriamycin injection. Scale bars: 100 μm ($\times 200$) (B).

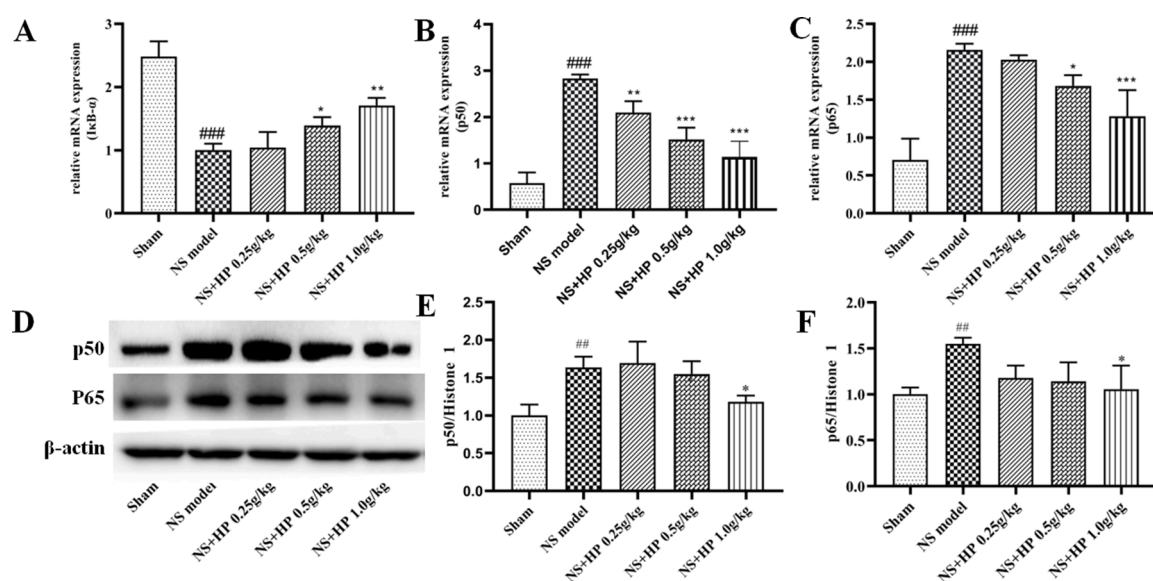


Fig. 4. Effect of HP on in NF- κ B pathways in NS rats. The mRNA expressions of *IkB- α* (A), *p50* (B), and *p65* (C) were determined in kidney tissue ($n = 6$). the protein levels of p50 and p65 in determined in kidney tissue ($n = 3$). Compared with sham group, # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$; Compared with NS group, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

“Component-target-pathway” ternary network construction for HP through network pharmacology

An integrated and classified network analysis was carried out to further explore the mechanism of action of HP. As shown in Fig. 7, 34 compounds were identified or tentatively characterized in HP. Their MS

information, including retention time, exact molecular weight, mass error, and characteristic fragment ions, is shown in Table 3 (structures are shown in Fig. S3). Among them, iridoids, flavonoids, saccharides, and organic acids were the main ingredients in HP aqueous solution. Compounds 9, 13, 14, 16, 17, 19, 30, and 31 were characteristic iridoids from *H. diffusa*, which is consistent with our previous research (Ye et al.,

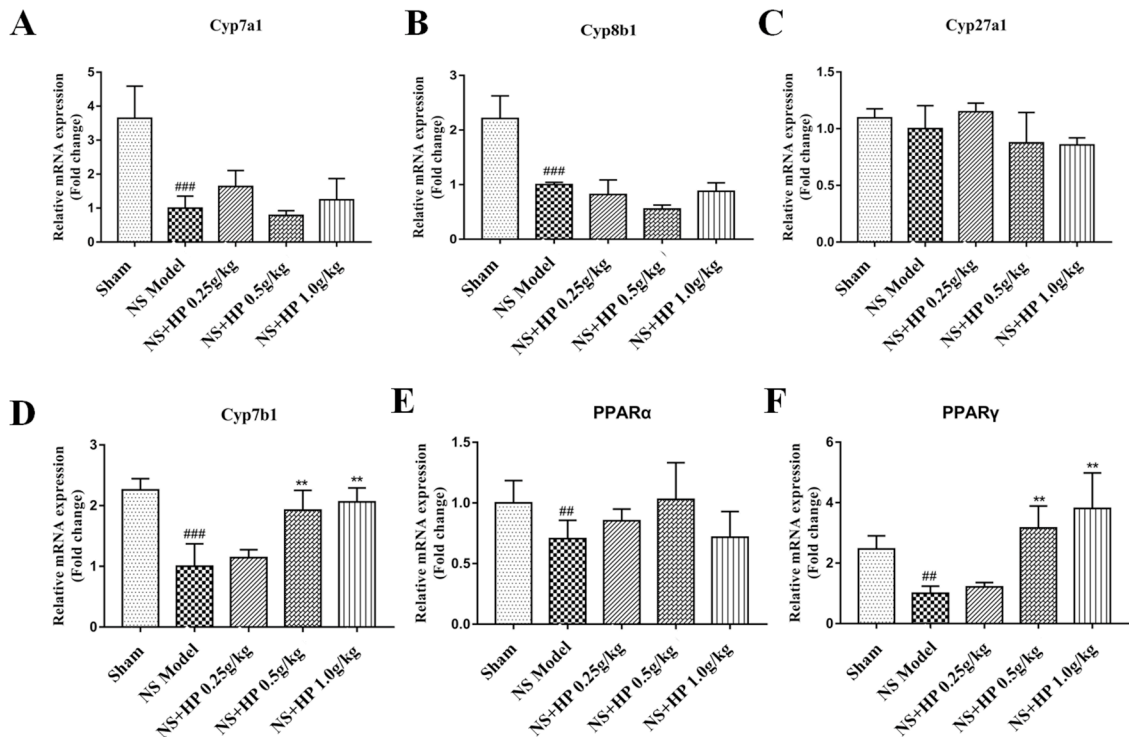


Fig. 5. Effect of HP on bile acid synthesis in NS rats. qPCR analysis of gene expression was performed for *Cyp7a1* (A), *Cyp8b1* (B), *Cyp27a1* (C), *Cyp7b1* (D), *PPARα* (E) and *PPARγ* (F) ($n = 6$). Compared with sham group, # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$; Compared with NS group, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

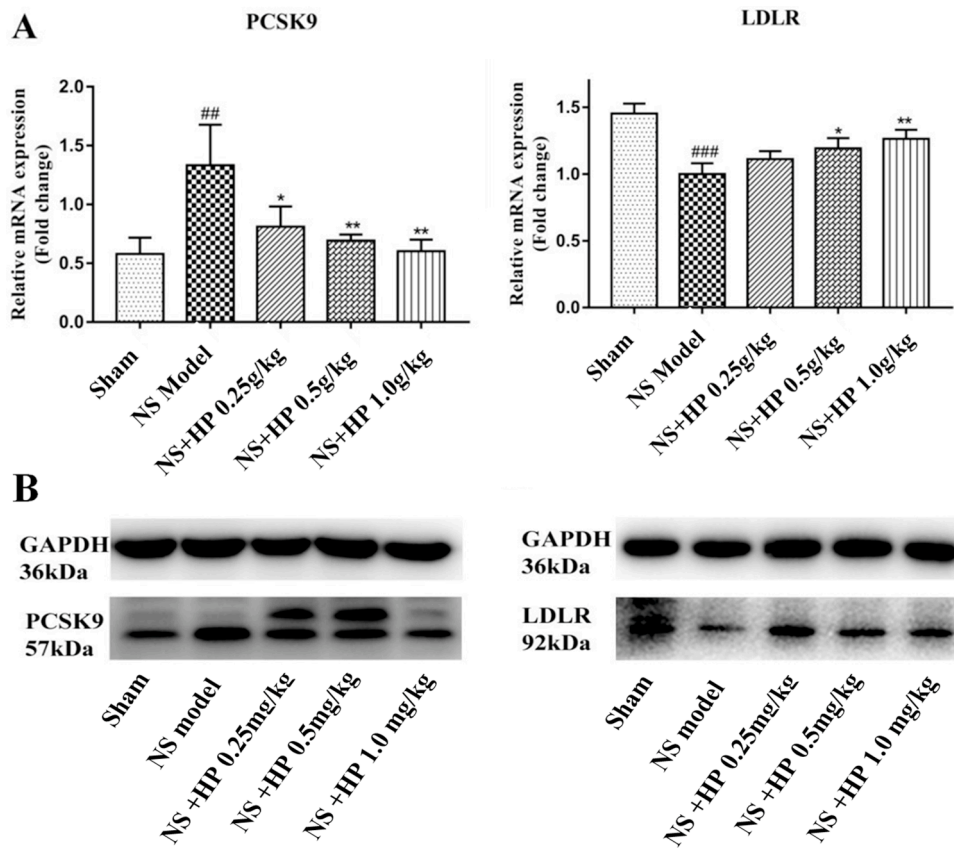


Fig. 6. Effect of HP on PCSK9-LDLR pathway in NS rats. A: the mRNA expressions of *PCSK9* and *LDLR* in liver tissue ($n = 6$); B: the protein levels of *PCSK9* and *LDLR* in liver tissue ($n = 3$). Compared with sham group, # $p < 0.05$, ## $p < 0.01$, ### $p < 0.001$; Compared with NS group, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

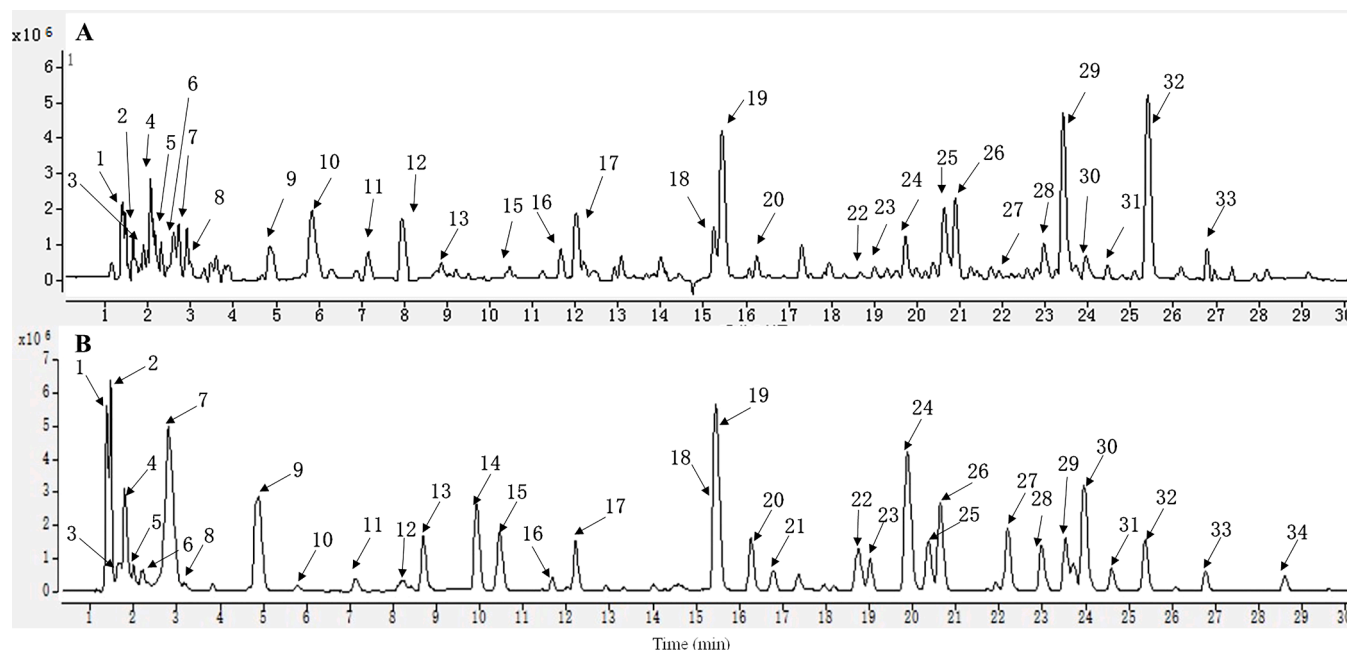


Fig. 7. The extracted ion chromatograms of 34 compounds of HP aqueous solution in positive ion mode (A) and negative ion mode (B).

2015). Compounds 5, 6, and 8 were saccharides that mainly existed in *I. rhizoma*. Compounds 25, 26, 28, 29, and 34 were flavonoids with quercetin aglycones. Deacetylasperulosidic acid and chlorogenic acid were usually used to monitor the qualities of *H. diffusa* and *I. rhizoma*, respectively (Zhai and Lv, 2016; Tong et al., 2020). Here, the contents of deacetylasperulosidic acid and chlorogenic acid in HP were $3648.96 \pm 30.94 \mu\text{g/g dw}$ and $66.99 \pm 0.44 \mu\text{g/g dw}$, respectively, as determined by the external standard method (Fig. S4 and Table S2).

Subsequently, the compound-target-network was constructed consisting of 82 nodes (23 compound nodes and 59 target nodes) and 461 interaction lines (Fig. 8A). Generally, one gene is targeted by multiple active compounds while one compound can target more than one gene, which reflects the characteristics of the interaction between multiple components and multiple targets for HP. The PPI network represented the protein interaction value with a score of > 0.7 (Fig. 8B). Inflammation-associated proteins (including ILs, RELA(p65), and NF- κ B1(p50) etc.) and lipid metabolism-associated proteins (including PCSK9, PPARA, PPARG, etc.) were the top 20 key targets, as shown in the bar chart (Fig. 8C), supporting the results of the animal experiment. GO biological process enrichment analysis was carried out according to $\text{FDR} < 0.05$ (Fig. 8D). HP is mainly involved in the regulation of cell death and apoptosis, the regulation of the inflammatory response, the regulation of the immune system, and the regulation of the lipid metabolism, etc., suggesting that HP plays a therapeutic role by regulating a variety of biological pathways. Through the KEGG pathway enrichment and analysis function of the DAVID platform, 15 signalling pathways, including NF- κ B, PPAR, etc., were screened according to $p < 0.001$ (Table S3, Fig. 8E). These pathways could interconnect and synergistically contribute to the therapeutic mechanisms of HP. Besides, the ternary network was established to provide chemical substances for subsequent mechanism verification.

Discussion

In recipes, *I. rhizoma* and *H. diffusa* are usually prescribed in a ratio of 1:1 with reference dosages of 9–30 g and 15–30 g, respectively (China Association of Chinese Medicine, 2011, 2019; Liu et al., 2010). In animal experiments, *I. rhizoma* showed an immunomodulatory effect at a dose of 0.5–2 g/kg (Wu et al., 2018); *H. diffusa* treatment of acute kidney

injury in mice was at 5 g/kg (Dai et al., 2017), and the protective dose of *H. diffusa* in the treatment of tumour rats was 3 g/kg (Wang et al., 2018). In consideration of the synergistic effects of the two herbs for long-term administration, the maximum administration dose of HP was designed to be 1.0 g/kg. Prednisone and atorvastatin were selected as positive drugs to investigate the anti-inflammatory and lipid-lowering effects of HP, respectively, and their doses were selected on the basis of clinical use (Fernandez-Juarez et al., 2018; Park et al., 2016). The adriamycin-induced nephropathy model is one of the most used experimental models of NS (Pereira Wde, et al., 2015). In this study, HP treatment for 4 weeks could improve renal function by reducing the expression of Upro and UA in a dose-dependent manner, relieve lipidaemia by reducing TC, TG, and LDL-C in serum, along with TG and LDL-C in renal tissue, and significantly decrease the expression of the inflammatory cytokines TNF- α and IL-1 β . All results showed a good effect of HP in the treatment of NS.

The NF- κ B signalling pathway is involved in a series of processes, such as the inflammatory response, immune response, oxidative stress, cell proliferation and apoptosis. The NF- κ B family includes p50, p52, Rel A (p65), Rel B, and c-Rel, of which the two subunits P50 and P65 are the most important (Zhang et al., 2017). TNF- α and IL-1 β are the early inflammatory mediators when the body undergoes a stress response, stimulating the cell surface receptors TNFR and IL-1R, respectively, and subsequently inducing the activation of NF- κ B. Moreover, the activation of NF- κ B can also enhance the transcription of the TNF- α and IL-1 β genes to form a vicious circle (Skaug et al., 2009). Previously, we proved that the water extract of *H. diffusa* showed a significant anti-inflammatory effect on lipopolysaccharide (LPS)-induced nephritis in mice. It significantly suppressed the production of TNF- α , IL-1 β , IL-6, and MCP-1 and significantly promoted the production of IL-10 in serum and renal tissues (Ye et al., 2015). Iridoids, including asperuloside, asperulosidic acid, desacetyl asperulosidic acid, scandoside methyl ester, scandoside, and *E*-6-*O*-coumaroyl scandoside methyl ester, were the main active components of *H. diffusa*, which showed different anti-inflammatory activities in LPS-induced RAW 264.7 cells via suppression of NF- κ B, p38, ERK, and JNK (He et al., 2018a, 2018b). In this paper, we still confirmed that HP treatment could significantly reduce the production of TNF- α and IL-1 β in a dose-dependent manner in both plasma and kidney tissue, and the mechanism of action is related to the inhibition of

Table 3

MS information of 34 compounds identified in HP.

No.	T _R (min)	Molecular Formula	[M-H] ⁺ (error, ppm)	Fragment ions	Identification	Source
1	1.376	C ₆ H ₁₂ O ₇	195.0518 (-3.95)	177[M-H-H ₂ O] ⁻ 159[M-H-2H ₂ O] ⁻	Gluconic acid	IR, HD
2	1.483	C ₆ H ₈ O ₇	191.0575 (-7.23)	173[M-H-H ₂ O] ⁻ 155[M-H-2H ₂ O] ⁻	Citric acid	IR, HD
3	1.606	C ₁₂ H ₂₂ O ₁₁	341.1103 (-3.99)	179[M-H-glc] ⁻	Cellobiose	IR, HD
4	1.801	C ₄ H ₆ O ₅	133.0146 (-2.64)	115[M-H-H ₂ O] ⁻	2-Hydroxybutanedioic acid	IR, HD
5	1.999	C ₁₈ H ₃₂ O ₁₆	503.1625 (-1.47)	341[M-H-glc] ⁻ 179[M-H-2glc] ⁻	Raffinose	IR
6	2.224	C ₂₄ H ₄₂ O ₂₂	665.2159 (-1.98)	503[M-H-glc] ⁻ 341[M-H-2glc] ⁻ 179[M-H-3glc] ⁻	Stachyose	IR
7	2.716	C ₅ H ₇ NO ₃	128.0345 (+6.33)	111[M-H-NH ₃] ⁻	Pyroglutamic acid	IR, HD
8	3.027	C ₃₀ H ₅₂ O ₂₆	827.2691 (-2.05)	647[M-H-glc-H ₂ O] ⁻ 503[M-H-2glc] ⁻ 341[M-H-3glc] ⁻ 179[M-H-4glc] ⁻	Maltopentaose	IR
9	4.879	C ₁₆ H ₂₂ O ₁₁	389.1099 (-2.47)	227[M-H-glc] ⁻	Deacetylasperulosidic acid	HD
10	5.791	C ₇ H ₆ O ₅	169.0146 (-2.08)	125[M-H-CO ₂] ⁻	Gallic acid	HD
11	7.134	C ₁₀ H ₁₃ N ₅ O ₅	282.0850 (-2.15)	150[M-H-C ₅ H ₈ O ₄] ⁻	Guanosine	IR, HD
12	8.157	C ₁₃ H ₁₆ O ₁₀	331.0678 (-2.20)	169[M-H-glc] ⁻	Galloyl-D-glucopyranose	IR
13	8.69	C ₁₇ H ₂₂ O ₁₂	417.1049 (-2.51)	371[M-H-HCOOH] ⁻	Deacetyl asperuloside	HD
14 [#]	9.925	C ₁₆ H ₂₂ O ₁₁	389.1098 (-2.22)	227[M-H-glc] ⁻ 209[M-H-glc-H ₂ O] ⁻	Scandoside	HD
15	10.454	C ₇ H ₆ O ₄	153.0198 (+2.16)	109[M-H-CO ₂] ⁻	3,4-Dihydroxybenzoic acid	HD
16	11.672	C ₁₇ H ₂₄ O ₁₁	403.1252 (-1.52)	241[M-H-glc] ⁻	Deacetyl asperulosidic acid methyl ester	HD
17 [#]	12.037	C ₁₇ H ₂₄ O ₁₁	403.1214 (+7.88)	241[M-H-glc] ⁻	Scandoside methyl ester	HD
18	15.444	C ₁₆ H ₁₈ O ₉	353.0886 (-2.24)	191[M-H-C ₆ H ₆ O ₃] ⁻	1-O-Caffeoylquinic acid	IR, HD
19 [#]	15.457	C ₁₈ H ₂₄ O ₁₂	431.1199 (-0.93)	371[M-H-CH ₃ COOH] ⁻ 269[M-H-glc] ⁻	Asperuloside acid	HD
20 [#]	16.256	C ₁₆ H ₁₈ O ₉	353.0883 (-1.40)	191[M-H-glc] ⁻	Chlorogenic acid	IR
21	16.754	C ₉ H ₈ O ₄	179.0352 (-1.21)	135[M-H-CO ₂] ⁻	Caffeic acid	IR, HD
22	18.761	C ₂₀ H ₂₀ O ₇	371.0990 (-1.69)	353[M-H-H ₂ O] ⁻ 175[M-H-C ₁₀ H ₁₂ O ₄] ⁻	Impeylenolide	IR
23	19.015	C ₁₇ H ₂₀ O ₉	367.1043 (-2.29)	193[M-H-C ₇ H ₁₀ O ₅] ⁻	3-O-Feruloylquinic acids	IR
24	19.878	C ₉ H ₈ O ₃	163.0402 (-0.81)	119[M-H-CO ₂] ⁻	p-Coumaric acid	IR, HD
25	20.384	C ₂₇ H ₃₀ O ₁₇	625.1360 (+8.02)	301[M-H-2glc] ⁻	Quercetin-3-O-sophoroside	HD
26	20.643	C ₂₇ H ₃₀ O ₁₇	625.1418 (-1.24)	301[M-H-gal-glc] ⁻	Quercetin-3-O-[2-O-D-glucopyranosyl]-β-D-galactopyranoside	HD
27	22.205	C ₂₁ H ₂₂ O ₈	401.1821 (-0.98)	357[M-H-HCOOH] ⁻ 221[M-H-C ₆ H ₁₂ O ₆] ⁻	Spionoside B	IR
28	22.987	C ₃₈ H ₄₀ O ₂₁	831.2002 (-1.51)	625[M-H-sinapol] ⁻ 445[M-H-sinapol-glc] ⁻ 303[M-H-sinapol-glc-gal] ⁻	Quercetin-3-O-[2-O-(6-O-E-sinapol)-β-D-glucopyranosyl]-D-galactopyranoside	HD
29	23.546	C ₃₈ H ₃₈ O ₂₀	801.1893 (+0.50)	625[M-H-feruloyl] ⁻ 300 [M-H-feruloyl-glc-gal] ⁻	Quercetin-3-O-[2-O-(6-O-E-feruloyl)-β-D-glucopyranosyl]-β-D-galactopyranoside	HD
30	23.964	C ₂₆ H ₃₀ O ₁₃	549.1567 (+8.48)	369[M-H-glc-H ₂ O] ⁻	6-O-Z-p-coumaroyl scandoside methyl ester	HD
31	24.607	C ₂₆ H ₃₀ O ₁₃	549.1585 (+5.21)	369[M-H-glc-H ₂ O] ⁻	6-O-E-p-coumaroyl Scandoside methyl ester	HD
32	25.373	C ₉ H ₁₆ O ₄	187.0980 (-2.22)	169[M-H-H ₂ O] ⁻ 143[M-H-CO ₂] ⁻ 125[M-H-2H ₂ O] ⁻	Azelaic acid	IR, HD
33 [#]	26.773	C ₂₇ H ₃₀ O ₁₆	609.1439 (+3.62)	301[M-H-rha-glc] ⁻	Rutin	IR, HD
34	28.621	C ₁₀ H ₁₈ O ₄	201.1135 (-1.32)	183[M-H-H ₂ O] ⁻ 139[M-H-H ₂ O-C ₂] ⁻	Sebacic acid	IR, HD

rha: rhamnose residue (C₆H₁₀O₄); glc: hexose residue (C₆H₁₀O₅); feruloyl: C₁₀H₈O₃; sinapol: C₁₁H₁₀O₄[#] Compound was identified by standard substance; IR: *Imperatae rhizoma*; HD: *Hedyotis diffusa*

p50 and p65.

As an important manifestation of NS, long-term hyperlipidaemia can not only increase cardiovascular disease risk, but also aggravate renal inflammation and kidney injury (Vaziri, 2016). The PPAR signalling pathway has been proven to play a critical role in lipid metabolism by regulating genes responsible for bile acid synthesis, such as cytochrome P450 (CYP) isoform 7a1 (CYP7a1), CYP27a1, and CYP8b1 (Ghonem et al., 2015). In this study, qPCR and western blot experiments showed that the relative expression levels of the metabolic enzymes CYP7a1, CYP7b1, and CYP8b1 in the liver of NS rats were significantly decreased, while the expression of CYP27a1 was not obviously changed. After HP oral administration for 4 weeks, the expression of CYP7b1 was upregulated, as was the upstream regulation of PPAR_γ rather than PPAR_α, suggesting that HP may regulate the bile acid synthesis pathway through

PPAR_γ-CYP7B1.

PCSK9, one of the top 20 targets, can promote LDLR degradation. LDLR is a class of membrane proteins richly expressed in hepatocytes, which can capture LDL particles as LDLR•LDL complex and import them into hepatic cells for destruction. However, the presence of PCSK9 can bind to LDLR, rearrange LDLR•LDL complexes, and inhibit recycling of LDLR, which results in an increased level of LDL in plasma (Stucchi et al., 2016; Taechalerpaisarn, et al., 2018). Hyperlipidaemia in NS has been proven to be closely related to the PCSK9-LDLR axis. In this study, the expression of LDLR mRNA in the NS model was significantly decreased, but the PCSK9 mRNA level was doubled compared with that in sham rats, which was consistent with previous reports (Liu and Vaziri, 2014). However, after oral administration of HP for 4 weeks, the expression of PCSK9 decreased and that of LDLR increased, suggested

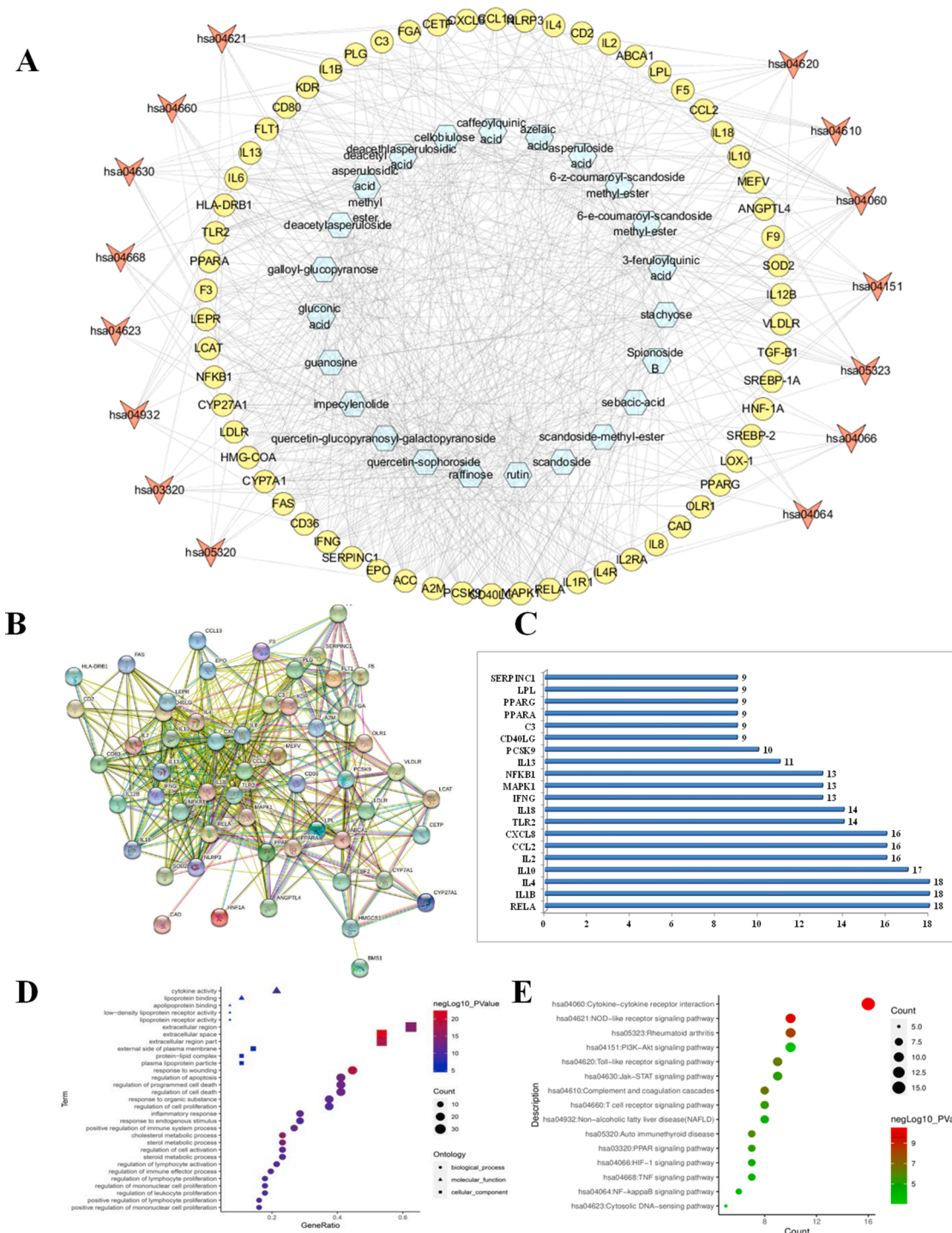


Fig. 8. The system network pharmacology analysis of HP on NS. A: the “compound-target-pathway” multivariate network of HP decoction. Component: blue hexagon; Target: yellow circle; pathway: red arrows. B: PPI network of HP-related targets (Nodes represent proteins; Node size decreases by degree value). C: Histogram of core target genes (Y axis is gene name; the number is the number of gene connection nodes. The more the number of gene connections, the more important the gene is, suggesting that the gene is more likely to be a key target gene for the treatment of NS). D: Bubble plot of GO functional enrichment (Y axis is the functional name of GO; X axis is the proportion of genes enriched in GO; the size of bubbles represents the number of genes contained; the redder the color, the smaller the p value). E: KEGG pathways for HP according to $p < 0.001$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

that this might be one of the hypolipidaemic mechanisms for HP.

Multicomponent analysis is the most prominent characteristic of traditional Chinese medicine, and it is difficult to clarify its molecular mechanism using a single “compound-target” model. Hence, a network pharmacological study was carried out. In the current database selection, the OMIM focuses on the genes related to genetic disease, the

DisGeNET puts its effort on the association between disease and inheritable/mutant gene, the MalaCards makes comprehensive information card for each disease, and the DigSee is a search engine in finding textual evidences to depict the gene which undergoes biological modification and leads to disease development. Therefore, these four databases could be complemented by each other and could offer a more comprehensive

target database for NS. Through this study, the “compound-target-pathway” multivariant network was established to further explore the mechanism of action of HP. In this “compound - target - pathway” diagram, PPAR γ (PPAR γ), PCSK9, RELA (p65), and NF- κ B1 (p50) were the top 20 targets for HP, supporting the previous findings of this study. However, other targets, such as Jak-STAT, PI3K-Akt, NOD-like receptor, MAPK, etc. mainly involved in anaphylaxis, organic lesion, cancer development, etc., may also be important for the HP treatment for NS and should be studied in further studies.

Conclusion

In the past few decades, NS has achieved a good clinical efficacy with glucocorticoids. However, NS often evolves into FRNS or SDNS, bringing great suffering to patients for long-term treatment, and some are SRNS ineffective glucocorticoids. HP was a clinical prescription for the treatment of kidney related diseases. In this study, we tried to explain the anti-inflammatory and hypolipidaemic effects of HP on NS and its mechanism. HP exhibited a significant anti-inflammatory effect by reducing the levels of inflammatory cytokines IL-1 β , TNF- α , MCP-1 and TGF- β 1 through the NF- κ B signalling pathway. Besides, HP also displayed a hypolipidaemic effect by decreasing TC, TG, and LDL-C, subsequently reducing lipid deposition in renal tissue. This hypolipidaemic effect might be related to the PPAR γ -CYP7B1 and CSK9-LDLR pathways. The results of the network pharmacology study supported these findings and offered new tasks to be carried out in the future.

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Author contributions

Lan Tang and Menghua Liu designed the study; Yaqian Dong, Shicong Yang, Lei La performed the animal experiment; Yaqian Dong, Wei Zou, Linna Gong, Yan Zhang, and Birui Shi performed the chemical analysis and network pharmacological study; Wei Zou, and Menghua Liu analyzed the data and wrote the paper. All data were generated in-house, and no paper mill was used. All authors agree to be accountable for all aspects of work ensuring integrity and accuracy.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.phymed.2021.153644](https://doi.org/10.1016/j.phymed.2021.153644).

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